Why CURRENT-MODE CONTROL In Switching Regulators Matters

There are thousands of different switching regulators on the market. Their election is based on specifications such as input voltage range, output voltage capability, maximum output current, and some other parameters. This article explains current mode, a differentiating feature commonly listed in data sheets, and its advantages and disadvantages

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The basic working principle of a current-mode regulator is shown in Fig. 1. Here, the feedback voltage is not only just compared with an internal voltage reference but also with a sawtooth voltage ramp for the generation of the necessary pulse width modulated (PWM) signal for the power switch.

The slope of this ramp is fixed in voltage-mode regulators. In current-mode regulators, the slope is dependent on the inductor current and is yielded from the current measurement shown in Fig. 1 at the switching node. This is what differentiates current-mode regulators from voltage-mode regulators.

Current-mode regulators offer many advantages. One is that the inductor current immediately adapts to changes in the input voltage (V_{IN} in Fig. 1). Thus, the input voltage change information is directly fed into the control loop, even before the output voltage (V_{OUT} in Fig. 1) tracks this input voltage change.

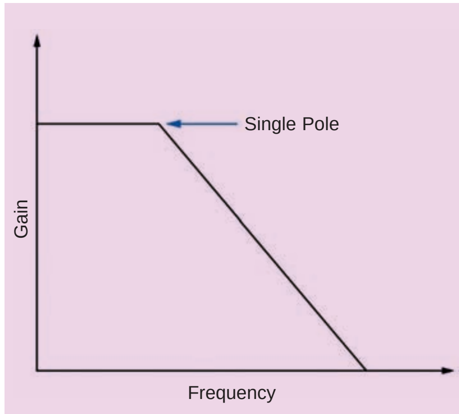

Fig. 2: A simplified control loop compensation through current-mode control shown in a Bode plot with just one simple pole in the power stage

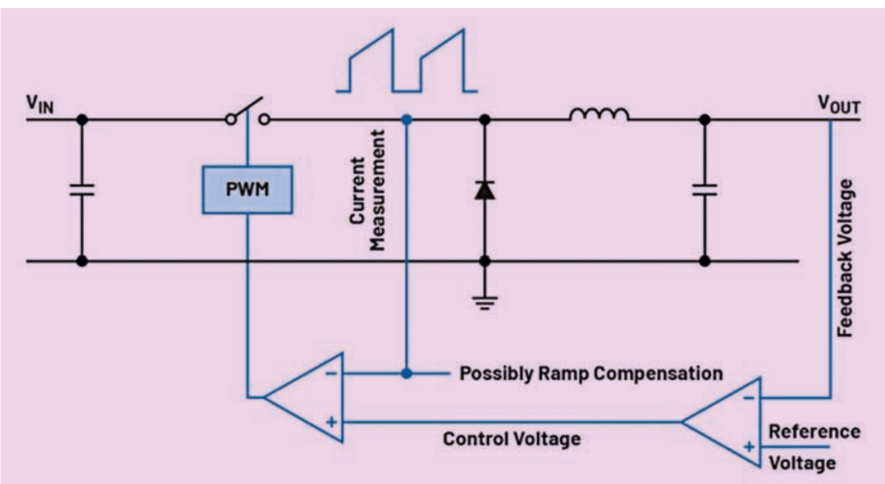

Fig. 1: The basic working principle of a current-mode regulator

The advantages of current-mode control are so convincing that most switching regulator ICs on the market work according to this current-mode control principle.

Another key advantage is simplified control loop compensation. The Bode plot of a voltage-mode regulator shows a double pole; a current-mode regulator generates just one simple pole of the power stage at this point. This produces a phase shift of 90°, instead of 180° with a double pole. Thus, a current-mode regulator can